Taniksha Datar

AI Engineer | LLM Systems · RAG · MLOps

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Education

Northeastern University **Expected Graduation: Dec 2026**

Master of Science in Computer Science *Boston, MA*

- GPA: 3.84 | Courses: DBMS, Algorithms, Artificial Intelligence, Cloud Computing, MLOps, NLP

Savitribai Phule Pune University **Jan 2020 – May 2024**

Bachelor of Engineering in Computer Engineering *Pune, India*

- GPA: 3.70 | Courses: Data Structures, Object Oriented Programming, Machine Learning, Computer Networks

Technical Skills

Languages: Python, SQL, Java, C++

ML & GenAI: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, Google ADK (multi-agent systems), MCP, agent orchestration, RAG, Vertex AI Search, Gemini, scikit-learn, FAISS, LLM evaluation, prompt engineering

MLOps & Cloud: GCP (Vertex AI, Cloud Run, Cloud Composer/Airflow, Cloud Build), AWS (EC2, S3), Docker, Terraform, MLflow, DVC, GitHub Actions, FastAPI, Snowflake, Snowflake Cortex, dbt

Data & Tools: Pandas, NumPy, Streamlit, Looker, Git, Selenium, BeautifulSoup

Work Experience

Bevi **Jan 2026 – Sept 2026**

Data Science Co-op *Boston, MA*

- Built an incremental dbt pipeline on Snowflake Cortex (Llama 3.1-405B) turning **100K+ tickets** and IoT alerts into repair histories for **36K+ machines**, served in Slack by its own **ADK agent** at **∼95%** lower cost and **10 min → 10 sec** lookup
- Shipped the **first knowledge retrieval agent** on the same Google ADK / Gemini 2.5 Flash **multi-agent architecture**, grounded in a Vertex AI Search index over internal SOPs with citations and abstention, now the **template for new agents**
- Modeled operational data as **Snowflake semantic views** served through a Snowflake agent, wrapped in a **custom MCP server** registered as a tool in the agentic architecture for **natural-language** queries over internal machine data
- Built a layered **evaluation harness** for the agent stack, pairing a retrieval regression suite with a **custom Vertex AI judge metric** that scores answer correctness on every release to catch regressions before they reach production

Reverie Language Technologies **Sept 2023 – Apr 2024**

Machine Learning Intern *Remote*

- Raised paraphrase quality up to **22% (ROUGE-L/BLEU/METEOR)** via domain-adaptive HuggingFace fine-tuning
- Curated **100K+ Hindi sentences** using semantic similarity and POS tagging to improve training data quality
- Cut training time **20%** by optimizing PyTorch training and batching; deployed via Docker, MLflow, GitHub Actions

Pune Institute of Computer Technology **Jan 2023 – Jun 2023**

Research Assistant *Pune, India*

- Built financial sentiment analysis models (SVM, LSTM, BERT), achieving **89% accuracy** and **86% F1-score**
- Scraped and processed **20K+ financial articles** with BeautifulSoup and Selenium into domain-specific NLP datasets
- Delivered the pipeline end-to-end (EDA, vectorization, training, deployment) via scikit-learn, TensorFlow, FastAPI

Projects

Break The Bot: LLM Safety Evaluation Pipeline | *MLOps/AI Safety* **GCP, Airflow, Groq, DVC, Terraform**

- Architected an automated LLM safety evaluation pipeline on Google Cloud Composer with Terraform-provisioned infrastructure and DVC-versioned datasets
- Implemented an LLM-as-a-Judge system using Groq (Llama-3.3) to score adversarial responses across **10+ models**
- Engineered strict data-quality gates with Great Expectations, giving the pipeline **fail-fast reliability** and drift detection
- Served metrics via a FastAPI service on Cloud Run (Cloud Build CI/CD) feeding a live Streamlit dashboard

Stack Overflow Auto-Tagging | *ML/NLP* **Python, HMM, Scikit-learn, Transformers**

- Engineered a Hidden Markov Model **from scratch** with Viterbi decoding, scoring **68.4% micro-F1** on 100K+ questions
- Benchmarked against Logistic Regression and DistilBERT baselines; fine-tuned DistilBERT reached **85% micro-F1**